given compound is organic or not. With the exception of few compounds like CCl_4, CS_2 all organic compounds also contain hydrogen. The presence of both these elements is confirmed by the following common test.

Copper(II)oxide test: The organic substance is mixed with about three times its weight of dry copper oxide by grinding. The mixture is then placed in a hard glass test tube fitted with a bent delivery tube. The other end of which is dipping into lime water in an another test tube. The mixture is heated strongly and the following reaction take place.

$$C + 2CuO \longrightarrow CO_2 + 2Cu$$

$$2H + CuO \longrightarrow H_2O + Cu$$

Thus if carbon is present, it is oxidized to CO_2 which turns lime water milky. If hydrogen is also present, it will be oxidized to water which condenses in small droplets on the cooler wall of the test tube and inside the bulb. Water is collected on anhydrous $CuSO_4$ which turns anhydrous $CuSO_4$ blue. This confirms the presence of C and H in the compound.

Detection of nitrogen by lassaigne sodium fusion test: This is a good test for the detection of nitrogen in all classes of nitrogenous compound and it involves the preparation of sodium fusion extract

This method involves the conversion of covalently bonded N, S or halogen present in the organic compounds to corresponding water soluble ions in the form of sodium salts For this purpose a small piece of Na dried by pressing between the folds of a filter paper is taken in a fusion tube and it is gently heated. When it melts to a shining globule, put a pinch of the organic compound on it. Heat the tube till reaction ceases and becomes red hot. Plunge it in about 50 mL of distilled water taken in a china dish and break the bottom of the tube by striking against the dish. Boil the contents of the dish for about 10 mts and filter. This filtrate is known as lassaignes extract or sodium fusion extract and it used for detection of nitrogen, sulfur and halogens present in organic compounds.

ii) Test for Nitrogen: If nitrogen is present it gets converted to sodium cyanide which reacts with freshly prepared ferrous sulphate and ferric ion followed by conc. HCl and gives a Prussian blue color or green colour or precipitate. It confirms the presence of nitrogen. HCl is added to dissolve the greenish precipitate of ferrous hydroxide produced by the excess of NaOH on $FeSO_4$ which would otherwise mark the Prussian blue precipitate. The following reaction takes part in the formation of Prussian blue.

$$Na + \underbrace{C + N}_{\text{from organic compounds}} \longrightarrow NaCN$$

$$FeSO_4 + 2NaOH \longrightarrow Fe(OH)_2 + Na_2SO_4$$

(from excess of sodium)

$$6NaCN + Fe(OH)_2 \longrightarrow \underset{\text{sod.ferrocyanide}}{Na_4[Fe(CN)_6]} + 2NaOH$$

$$3Na_4[Fe(CN)_6] + 4FeCl_3 \longrightarrow \underset{\substack{\text{ferric ferrocyanide} \\ \text{Prussian blue or green ppt}}}{Fe_4[Fe(CN)_6]_3} + 12\,NaCl$$

Incase if both N & S are present, a blood red colour is obtained due to the following reactions.

Unit 11 revised - jagan.indd 138 22-01-2021 12:42:27